

Regulated Radial L^p Multiplier Profiles Have Continuous Representatives

Candidate partial result for arXiv:2602.14623

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Status. Candidate partial result, likely valid. This note does not solve the full radial multiplier continuity question.

1 Source question

Mikael de la Salle asks the following in Question 1.6 of [1], page 4:

Is there a discontinuous function $m : (0, \infty) \rightarrow \mathbb{R}$ such that the Fourier multiplier with symbol $\xi \in \mathbb{R}^d \mapsto m(|\xi|)$ is bounded on $L_p(\mathbb{R}^d)$ for some $p \neq 2$?

Figure 1 is a crop of the original source page.

to the "function" whose Fourier transform is the pointwise multiplication $\varphi \hat{f}$, for some function $\varphi : \mathbf{R}^d \rightarrow \mathbf{C}$ called the symbol of the Fourier multiplier. It is said to L_p -bounded if it maps $L_p(\mathbf{R}^d)$ to itself, and radial if $\varphi(\xi)$ only depends on the Euclidean norm $|\xi|$. Here and throughout the paper we use the convention $\hat{f}(\xi) = \int f(x) e^{-2i\pi \langle x, \xi \rangle} dx$ for the Fourier transform.

This motivates, as a baby case of Question [1.5](#) the following question. It is very natural from the non-commutative perspective, but it does not seem to have been extensively studied.

Question 1.6. Is there a discontinuous function $m : (0, \infty) \rightarrow \mathbf{R}$ such that the Fourier multiplier with symbol $\xi \in \mathbf{R}^d \mapsto m(|\xi|)$ is bounded on $L_p(\mathbf{R}^d)$ for some $p \neq 2$?

Figure 1: Question 1.6 in [1, p. 4].

The source proves that if the multiplier is bounded, then the primitive of $t \mapsto m(e^t)$ is smooth in Zygmund's sense [1, Theorem 1.7]. Proposition 2.4 of the same paper gives the equivalent weak-* tangent formulation used below.

2 Definitions

A function $u : I \rightarrow \mathbb{R}$ on an open interval is *locally regulated* if at every $t \in I$ the finite one-sided limits $u(t-)$ and $u(t+)$ exist. We do not require either one-sided limit to equal the displayed point value $u(t)$.

Two bounded symbols represent the same Fourier multiplier when they agree almost everywhere. Thus the appropriate continuity conclusion is the existence of a continuous representative of the multiplier symbol.

3 Idea of the proof

The source theorem excludes every nonconstant weak-* tangent of the logarithmic radial profile. A regulated jump has a canonical tangent: zooming at the jump converges to the step function whose two values are the left and right limits. The source theorem therefore forces those limits to agree at every point. What remains are only removable point-value defects. A compact interval can contain only finitely many defects of any fixed size, so their union is countable and replacing them by the common two-sided limits produces a continuous representative.

The highly oscillatory symbol singled out in Question 1.8 is not regulated near the unit sphere, so this mechanism deliberately does not address it.

4 Partial theorem and proof

Theorem 1. *Let $d \geq 2$, $1 < p < \infty$, and $p \neq 2$. Let $m \in L_\infty(0, \infty; \mathbb{R})$ and assume that*

$$\widehat{T_m f}(\xi) = m(|\xi|)\widehat{f}(\xi)$$

defines a bounded operator on $L_p(\mathbb{R}^d)$. If the equivalence class of m contains a locally regulated representative, then it contains a continuous representative on $(0, \infty)$.

Proof. Choose a locally regulated representative of m . Composing it with the continuous clipping map to an essential L_∞ bound changes it only on a null set and preserves regulatedness. We may therefore take this representative to be bounded; still denote it by m , and put

$$u(t) = m(e^t), \quad t \in \mathbb{R}.$$

Then u is bounded and locally regulated. By Theorem 1.7 and Proposition 2.4 of [1], every weak-* limit in $\sigma(L_\infty, L_1)$ of a sequence

$$x \mapsto u(s_n + x/r_n), \quad r_n \rightarrow \infty,$$

is constant almost everywhere.

Fix $s \in \mathbb{R}$ and set $a = u(s-)$ and $b = u(s+)$. Take $s_n = s$ and $r_n = n$. For each fixed $x < 0$,

$$u(s + x/n) \rightarrow a,$$

and for each fixed $x > 0$ the same expression converges to b . The functions are uniformly bounded, so dominated convergence against an arbitrary $L_1(\mathbb{R})$ test function shows weak-* convergence to

$$v(x) = a\mathbf{1}_{(-\infty, 0)}(x) + b\mathbf{1}_{(0, \infty)}(x).$$

The tangent characterization forces v to be constant almost everywhere; hence $a = b$. Since s was arbitrary, the two one-sided limits of u agree at every point. Denote their common value by $c(s)$.

We next verify that c is continuous. Given s and $\varepsilon > 0$, the one-sided limit definitions give $\delta > 0$ such that

$$|u(t) - c(s)| < \varepsilon \quad \text{whenever} \quad 0 < |t - s| < \delta.$$

If $s < t < s + \delta$, then points approaching t from either side may be chosen inside $(s, s + \delta)$, and passing to the corresponding one-sided limit gives $|c(t) - c(s)| \leq \varepsilon$. The same argument applies on the left. Thus c is continuous.

It remains to show that $u = c$ almost everywhere. For integers $N, k \geq 1$, set

$$E_{N,k} = \{s \in [-N, N] : |u(s) - c(s)| \geq 1/k\}.$$

This set is finite. Indeed, if it had an accumulation point s_0 , choose distinct $s_j \in E_{N,k}$ with $s_j \rightarrow s_0$ and pass to a one-sided subsequence. Regulatedness gives $u(s_j) \rightarrow c(s_0)$, while continuity gives $c(s_j) \rightarrow c(s_0)$, contradicting $|u(s_j) - c(s_j)| \geq 1/k$. Therefore $\{u \neq c\} = \bigcup_{N,k} E_{N,k}$ is countable, and $u = c$ almost everywhere.

Finally define $\tilde{m}(r) = c(\log r)$. This function is continuous on $(0, \infty)$ and equals $m(r)$ almost everywhere, so it represents the same Fourier multiplier. $\square$

Corollary 2. *Under the multiplier assumptions of Theorem 1, the conclusion holds whenever m is locally of bounded variation, locally of finite q -variation for some $1 \leq q < \infty$, or locally cadlag. In the cadlag case, m itself (rather than merely an almost-everywhere representative) is continuous.*

Proof. Each listed hypothesis implies local regulatedness. For finite q -variation, failure of a one-sided limit would allow two interlaced sequences with values separated by a fixed positive amount, forcing the sum of the q th powers of successive increments to be infinite. Theorem 1 applies. A cadlag function already equals its right-hand limit; the proof of the theorem forces the left-hand limit to have the same value, so the function is continuous at every point. $\square$

5 Verification notes

- The proof uses no unproved lemma. Its only external dependency is the weak-* tangent characterization in [1, Theorem 1.7 and Proposition 2.4].
- The step-function tangent is obtained in the full $\sigma(L_\infty, L_1)$ topology, not merely pointwise locally: dominated convergence applies to every $L_1(\mathbb{R})$ test function.
- The representative issue is explicit. The proof does not claim that arbitrary null-set changes of a multiplier symbol are continuous.
- No computation is used or needed.

Verifier verdict: likely valid. The most useful human check is the elementary compactness argument proving that $u = c$ almost everywhere.

6 Scope and novelty status

The full Question 1.6 remains open for bounded measurable profiles, and the specific slowly oscillating profile in Question 1.8 is outside the regulated class. The result is therefore only a regularity-class subcase.

On 11 August 2026, the local run indexes and parsed arXiv corpus were searched for the exact arXiv id and close terminology. Bounded arXiv/web searches used the exact id, the exact slowly oscillating candidate, and combinations of “regulated”, “one-sided limits”, “bounded variation”, and “radial Fourier multiplier”. No explicit prior statement of this corollary was found. Novelty confidence is modest because the proof is an elementary deduction from Proposition 2.4 of the source paper itself.

7 Human review recommendation

Check the regulated-function lemma and retain the packet, if accepted, as a narrow partial result. It should not be described as resolving the general continuity question or Question 1.8.

References

- [1] Mikael de la Salle, *Kekeya conjecture and High-Rank Lattice von Neumann algebras*, arXiv:2602.14623v1, 2026. <https://arxiv.org/abs/2602.14623>